

LA Crime Analytics Dashboard Application

(User Manual)

1. Introduction

The LA Crime Analytics Dashboard is a standalone desktop application that visualizes crime patterns in Los Angeles.

Built with R Shiny and packaged using Electron, this application runs without requiring R or RStudio installation.

Key Features

- no R installation is required
- works as a standalone .exe
- interactive crime map with clustering
- 32 dynamic visualization types
- filtering by year, area, crime type, and hour
- Export charts as PNG images

2. Installation & Setup

Running the Application (No R Installation Required) :-

- a) Download the (LA Crime Analytics Dashboard-win32-x64-0.0.1.zip) folder from the Dropbox link
- b) after downloading the file , The file is compressed (ZIP format). You need to extract it:

- Right-click the downloaded ZIP file
- Select "Extract All"
- Choose a destination folder (e.g., Desktop or Documents)
- Click "Extract"

Note: The dataset is large (206MB). More ram is recommended for smooth performance. Extraction may take 1-2 minutes.

c) launching the Application

- Navigate to the extracted file location
- Double-click (LA Crime Analytics Dashboard.exe)
- A console window will open (don't close it)
- Wait for the application to load (~1-2 minutes)
- The dashboard will show all plots and be ready for use

3. Dashboard Overview

a) Top Section - Key Metrics (4 Value Boxes)

Box	Description
Total Crimes	Total incidents matching current filters
Violent Crimes	Assault, homicide, rape, robbery
Property Crimes	Burglary, theft, shoplifting
Arrest Rate	Percentage of incidents resulting in arrest

b) Left Sidebar - Filter Panel

Contains all filtering options (filtering by year, area, crime type, and hour)

c) Main Content Area

1- Dashboard Tab

	Content	Description
1	Interactive Map + Top Crime Types	Leaflet map with crime clustering + bar chart of most common offenses
2	Top Areas by Crime Count	Ranked bar chart showing which police divisions have highest crime volume
3	Monthly Trends + Hourly Distribution	Seasonal patterns by month + peak crime hours throughout the day
4	Category Pie Chart + Area Treemap	Crime category proportion + hierarchical view of crimes per area
5	Victim Sex + Age Groups + Day/Hour Heatmap	Demographics of victims + heatmap of busiest day/hour combinations
6	Yearly Trends + Top Weapons	Long-term crime trends + most common weapons used
7	Arrest Rate by Area + Arrest Rate by Hour	Enforcement effectiveness across police divisions + by time of day
8	Top Premise Types + Category by Premise	Where crimes occur (street, residence, vehicle, etc.) + breakdown by crime type
9	Age Distribution per category (Violin Plot)	Distribution of victim ages for each crime type (shows shape, median, outliers)
10	Crime Flow (Parallel Sets)	Interactive flow diagram: Crime Category → Victim Sex → Victim Descent

2- Advanced Analysis Tab

	Content	Description
1	LOESS Smoothed Crime Trend	Monthly trends with 95% confidence bands (removes seasonal noise to show true direction)
2	Scree Plot + PCA Biplot	Principal Component Analysis: variance explained + crime pattern dimensions with variable loadings
3	Correlogram + Arrest Rate Error Bars	Correlation heatmap of numeric features + arrest rate per area with 95% confidence intervals
4	Crime Cartogram Heatmap	Equal-area tile map of 21 police divisions (toggle between Crime Count and Arrest Rate)
5	Small Multiples	Hour-of-day crime profile for EVERY police area (21 mini bar charts with peak hour annotation)
6	Ridgeline Plot	Hour-of-day distribution comparison across crime categories (density ridges)
7	2D Density Map	Victim age vs. hour-of-day heatmap (one panel per crime category)
8	Average Victim Age by Hour (Bubble Plot)	Bubble chart showing avg victim age per hour + LOESS trend line + overall average
9	Connected Scatter (Age vs. Crime Count)	Crime count across every victim age (0-100) with peak age annotations per category
10	Confidence Strip	Mean victim age per area with 80/95/99% confidence intervals (shows statistical uncertainty)
11	Arrest Rate by Crime Category	Bar chart showing arrest rate for each crime type with overall average line
12	Weekend vs. Weekday Crime Profile	Daily average comparison between weekends and weekdays

d) Using the Filters (Left Sidebar)

Filter	Purpose	How to Use
Year	Select specific year	Dropdown - choose "All" or specific year
Area	Focus on specific police area	Dropdown - select "All Areas" or district
Crime Category	Filter by crime type	Check/uncheck boxes
Hour of Day	Time range filter	Drag slider handles (0-23)

e) Accessibility Options (Color Vision Modes)

Located below the Hour slider in the left sidebar, the **Color Vision Mode** selector makes all charts accessible to users with color vision deficiencies.

Mode	Best For	Color Changes
Normal	Standard color vision	Default red/orange/blue/purple palette
Deuteranopia	Red-green color blindness (most common)	Reds become orange/brown, greens become yellow/tan
Protanopia	Red-green color blindness	Reds shift to dark brown, greens to light yellow
Tritanopia	Blue-yellow color blindness (rare)	Blues become teal, yellows become pink

Filter Examples

Find nighttime violent crimes:

- Category: Check only "Violent Crime"
- Hour: Drag to 20-23 (8 PM - 11 PM)

Focus on one area in 2023:

- Year: Select "2023"
- Area: Select specific district (e.g., "Central")
- Keep other filters as is

Reset all filters:

- Year → "All"
- Area → "All Areas"
- Check all Category boxes
- Hour → 0 to 23

4) Visualization Guide

Interactive Map

- Click marker → View crime details (type, date, location)
- Click cluster → Zoom in to see individual crimes
- Scroll → Zoom in/out
- Drag → Pan around map

All Charts

Action	Result
Hover	See exact values
Camera icon	Download as PNG
Zoom in / Zoom out	Zoom in or out of plot
Box Select/Lasso Select	Box/lasso select area
Home icon / Autoscale	Reset view
Double-left click	Reset zoom
Change Color Vision Mode	Sidebar dropdown → All charts update automatically

→ All charts update automatically

Chart-Specific Tips

Chart	What to Look For
Top Crime Types	Most common offenses
Monthly Trends	Seasonal patterns
Hourly Distribution	Peak crime times
Heatmap	Busiest day/hour combinations
Arrest Rate by Area	Which areas have highest/lowest arrest rates
Yearly Trends	Crime increasing or decreasing over time

5) Quick Reference: Which Chart Should I Use?

I want to know	Best Chart	Tab
Where are crimes located?	Interactive Map	Dashboard
Which area has most crimes?	Top Areas by Crime Count	Dashboard
Is crime seasonal?	Monthly Trends	Dashboard
What time is most dangerous?	Hourly Distribution	Dashboard

I want to know	Best Chart	Tab
Is crime really trending up?	LOESS Smoothed Trend	Advanced Analysis
What drives crime patterns?	PCA Biplot + Scree	Advanced Analysis
Which areas need attention?	Cartogram (toggle Arrest Rate)	Advanced Analysis
Can I trust this area's data?	Confidence Strip	Advanced Analysis
When do different crimes happen?	Ridgeline Plot	Advanced Analysis
How does victim age vary by hour?	Bubble Plot (Avg Age by Hour)	Advanced Analysis
Compare all 21 areas at once?	Small Multiples	Advanced Analysis

I want to know	Best Chart	Tab
Is enforcement effective?	Arrest Rate by Category	Advanced Analysis
Do weekends need different patrol?	Weekend vs. Weekday	Advanced Analysis

6) Quick Fixes

- **Reset all filters** - Fixes most "no data" issues
- **Restart the app** - Fixes most crash issues
- **PCA plots are empty** - Need at least 50 crimes after filtering (widen your year/area/hour filters)
- **Cartogram missing squares** - Some areas have zero crimes with current filters (normal behavior)
- **Colors look different than expected** - Check Color Vision Mode in sidebar - change to "Normal"
- **Confidence strips show wide intervals** - That area has few crimes treat estimate with caution (this is normal)